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Damage characteristics and constitutive modeling of the 2D C/SiC composite: Part II – Material model and numerical implementation

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Abstract In this work, a macroscopic non-linear constitutive model accounting for damage, inelastic strain and unilateral behavior is proposed for the 2D plain-woven C/SiC composite. A set of scalar damage variables and a new thermodynamic potential expression are introduced in the framework of continuum damage mechanics. In the deduced constitutive equations, the material's progressive damage deactivation behavior during the compression loading is described by a continuous function, and different deactivation rates under uniaxial and biaxial compression loadings are also considered. In damage evolution laws, the coupling effect among the damage modes and impediment effect of compression stress on the development of shear damage in different plane stress states are taken into account. Besides, the general plasticity theory is applied to describing the evolution of inelastic strain in tension and/or shear stress state. The Tsai–Wu failure criterion is adopted for strength analysis. Additionally, the material model is implemented as a user-defined material subroutine (UMAT) and linked to the ABAQUS finite element software, and its performance is demonstrated through several numerical examples.

1. Introduction

As a typical thermo-structural material, the mechanical behavior and strength of the C/SiC composite and structures are of great concern to researchers. Considering the main drawback of the high fabrication cost, it is meaningful to develop an appropriate constitutive model for this material which is convenient for integration into numerical tools such as the finite element (FE) software, since FE analysis of composite structures can replace some mechanical tests and provide the engineers with the necessary information to achieve optimized structural design.[1] However, it is not an easy task to achieve this goal[2] due to the complex damage mechanisms and the resultant non-linear mechanical behavior of this material.[3]

Phenomenological continuum damage mechanics (CDM) models are generally based on the thermodynamics of irreversible processes,[4] in which the material damage is represented by internal variables. Although they cannot explicitly account for

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microscopic features like the microstructural evolution,[5] the CDM models are suitable for structural analysis and the model parameters can be determined by macroscopic experiments.[6] Furthermore, they can be combined with the general plastic theory to describe the irreversible deformation of some materials. In view of these advantages, the CDM approach is attractive to researchers when modeling the mechanical behavior of ceramic matrix composites (CMCs), since damage is the main source of energy dissipation in them.

Researchers from the well-known LMT[7–11] and ONERA[2,12–14] laboratories have done a systematic work concerning the development of the CDM models for CMCs. In the representative "LMT" models, Ladevèze and Gasser et al. have distinguished matrix and fiber-yarn damage mechanisms of the SiC/SiC composite,[7,8] and isotropic microcraking and directional damage are separately taken into account in each damage mechanism. Meanwhile, the unilateral behavior is considered by dividing the Gibb's free energy into "tension" and "compression" parts, and a spectral decomposition of the stress tensor is generally needed.[7–9] These models have been widely applied to CMCs including SiC/SiC and C/C composites.[8,9] Recently, Cluzel and Genet et al. have extended the applications of the "LMT" models to self-healing CMCs by coupling the anisotropic macroscopic damage behavior with physicochemical mechanisms,[10,11] and both the mechanical behavior and lifetime of the CMC structures subjected to complex thermo-mechanical loadings in oxidizing environments can be predicted by their models.

Chaboche and Maire et al. have developed different CDM models which are referred to as "ONERA" models.[2,12] Typically, scalar and tensorial damage variables were used to describe the microcracks oriented respectively by the reinforcements and the applied load in these models. The thermal residual stress (or strain) and damage deactivation indices are incorporated in the formulation of thermodynamic potential either in Gibbs or Helmholtz form. A new damage deactivation rule in Ref.[13] allows a better description of the progressive deactivation effect and the residual strain. The "ONERA" models have been applied to the C/SiC composite[12]; however, the calibration of constitutive equations and the FE implementation are difficult jobs as these models are highly complicated.

In recent years, some researchers have proposed simplified macroscopic or multiscale material models for CMCs which are more convenient to achieve their engineering applications.[14–19] However, little work has been especially devoted to the development of macroscopic constitutive models for the C/SiC composite. The present paper is to develop a macroscopic material model for the 2D plain-woven C/SiC composite, in which the evolution of damage and inelastic strain as well as the progressive damage deactivation behavior will be considered. An integration algorithm and a consistent tangent stiffness matrix based on the material model are provided to achieve its FE implementation. In accordance with the first part of the paper,[3] the model is confined to in-plane static loading conditions at ambient temperature.

2. Material model

Firstly, let us briefly summarize the damage characteristics of the material under in-plane static mechanical loadings as observed in the experiment,[3] which include

(1) The material shows remarkable non-linear stress/strain relationships due to the initiation and development of damage and inelastic strain when subjected to tension and/or shear loading condition.
(2) Compared with the uniaxial tension and pure shear loading conditions, the combined tension and shear loads accelerate the growth rates of damage variables on the material's principle axes, and contrarily the compression stress partially impedes the shear damage evolution under combined compression and shear loadings. The development of inelastic strain shows a similar evolution rule.
(3) The material exhibits a nearly linear-elastic behavior under monotonic uniaxial compression. Besides, the tension and/or shear damage generated in the loading history will be gradually deactivated during compression, and the biaxial compression stresses yield a faster damage deactivation rate than the uniaxial compression condition.

Considering these mechanical responses, the material model will be developed based on the CDM and plastic theories.

2.1. Stress/strain relationships

The microscopic observations have revealed the crack networks perpendicular to the reinforcing fiber bundles,[3] which are probably caused by the applied tension stress superposed on the tensile thermal residual stress in the matrix. The interface debonding flaws are naturally aligned with the fibers. Indeed, some matrix cracks orthogonal to the loading directions are also observable in the material, which belong to the isotropic damage mode. In the present paper, however, we will only consider the orthotropic damage modes for the sake of simplicity, and accordingly the orthotropic symmetry of the material is assumed to be maintained throughout the damage process under infinitesimal deformation. Moreover, the compliance coefficient $-\nu_{12}/E_1$ is assumed to be constant.[20] Based on these hypotheses, we introduced a set of in-plane scalar damage variables: d_1, d_2, d_6 ($0 \leqslant d_{i=1,2,6} \leqslant 1$) in the material coordinate system with similar definitions in Ref.[3]:

$$\begin{cases} d_1 = 1 - \frac{E_1}{E_1^0} \\ d_2 = 1 - \frac{E_2}{E_2^0} \\ d_6 = 1 - \frac{G_{12}}{G_{12}^0} \end{cases} \tag{1}$$

where E_1, E_2 and G_{12} are damaged elastic moduli of the material under mechanical loadings, and the superscript 0 denotes the initial state of the material.

Referring to the formulations in the "LMT" models concerning the damage and unilateral behavior of the SiC/SiC, SiC/MAS-L and C/C composites,[7,21,22] we divided the complementary elastic strain energy density $\rho\psi_e$ into several terms, where ρ is the material density, ψ_e the complementary elastic strain energy. However, unlike the damage characteristics of the SiC/SiC composite which show a nearly instant stiffness recovery after the load reversal from tension to compression[7] and the damageable behavior of the C/C composite under compression loadings,[20] the compression stress causes no apparent damage in the 2D plain-woven C/SiC material. Instead, it would gradually deactivate the damage formed in the loading history (if any) by closing the microcracks and interface debonding flaws. Therefore, we formulate $\rho\psi_e$ of the damaged material as follows:

$$\rho\psi_e = \frac{1}{2}\left[\frac{\langle\sigma_1\rangle^2}{E_1^0(1-d_1)} + \frac{\varphi(\langle-\sigma_1\rangle, d_1)}{E_1^0} - \frac{2\nu_{12}^0\sigma_1\sigma_2}{E_1^0} + \frac{\langle\sigma_2\rangle^2}{E_2^0(1-d_2)} + \frac{\varphi(\langle-\sigma_2\rangle, d_2)}{E_2^0} + \frac{\sigma_6^2}{G_{12}^0(1-d_6)}\right] \tag{2}$$

where σ_1, σ_2 and σ_6 are components of Cauchy nominal stress tensor $\boldsymbol{\sigma}$, ν_{12}^0 is the primary in-plane Poisson's ratio of the undamaged material; $\langle\rangle$ denotes the McAuley operator with a definition of $\langle x\rangle = (x + |x|)/2$; the terms related to $\varphi(\langle-\sigma_i\rangle, d_i)$ represent the strain energy generated during the monotonic compression ($d_i = 0$) or the damage deactivation process ($d_i > 0$) ($i = 1, 2$), and the function φ is defined as

$$\varphi(\langle-\sigma_i\rangle, d_i) = \frac{1}{k_i^2}\left[\langle-k_i\sigma_i\rangle^2 + \ln\left(1 - d_i e^{-\langle-k_i\sigma_i\rangle^2}\right) - \ln(1-d_i)\right] \quad (i=1,2) \tag{3}$$

where no summation is assumed over a repeated index (the same setting is adopted in the rest of the paper), and k_i (MPa^{-1}) is a non-negative parameter related to the thermal residual stress (σ_r) along the fiber direction and defined as

$$k_i = \frac{\alpha_i}{\sigma_r} \quad (i=1,2) \tag{4}$$

where α_i is a damage deactivation parameter on the ith principle axis of the material.

In order to differentiate the damage deactivation rates caused by the uniaxial and biaxial compression stress states, we define the damage deactivation parameters as

$$\begin{cases} \alpha_1 = g(\sigma_1)\alpha_{11} + g(\sigma_2)\alpha_{12} \\ \alpha_2 = g(\sigma_2)\alpha_{22} + g(\sigma_1)\alpha_{21} \end{cases} \tag{5}$$

where the material coefficients α_{ij} (i, j = 1, 2) represent the deactivation effect of compression stress σ_j on the tensile damage on the ith principle axis of the material, and according to the symmetry of plain-woven fabrics, there are $\alpha_{11} = \alpha_{22}$ and $\alpha_{12} = \alpha_{21}$; $g(x)$ is function slightly different from the Heaviside function and defined as

$$g(x) = \begin{cases} 0 & \text{if } x \geqslant 0 \\ 1 & \text{if } x < 0 \end{cases} \tag{6}$$

From Eqs. (2) and (6), one can determine that $\rho\psi_e$ is positive definite (the proof is presented in the Appendix A) and equal to zero when $\boldsymbol{\sigma} = \mathbf{0}$. The elastic strain tensor is equal to the derivative of $\rho\psi_e$ with respect to the stress tensor $\boldsymbol{\sigma}$:

$$\boldsymbol{\varepsilon}^e = \rho\frac{\partial\psi_e}{\partial\boldsymbol{\sigma}} = (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p) \tag{7}$$

where $\boldsymbol{\varepsilon}^e$ and $\boldsymbol{\varepsilon}^p$ denote the elastic and inelastic parts of the strain tensor $\boldsymbol{\varepsilon}$, respectively.

Substituting Eqs. (2) and (3) into the above equation, the damage elastic law is derived:

$$\boldsymbol{\varepsilon}^e = \begin{bmatrix} \varepsilon_1^e \\ \varepsilon_2^e \\ \varepsilon_6^e \end{bmatrix} = \begin{bmatrix} \frac{\langle\sigma_1\rangle}{E_1^0(1-d_1)} - \frac{\langle-\sigma_1\rangle}{E_1^0\left(1-d_1 e^{-\langle-k_1\sigma_1\rangle^2}\right)} - \frac{\nu_{12}^0}{E_1^0}\sigma_2 \\ \frac{\langle\sigma_2\rangle}{E_2^0(1-d_2)} - \frac{\langle-\sigma_2\rangle}{E_2^0\left(1-d_2 e^{-\langle-k_2\sigma_2\rangle^2}\right)} - \frac{\nu_{12}^0}{E_1^0}\sigma_1 \\ \frac{\sigma_6}{G_{12}^0(1-d_6)} \end{bmatrix} \tag{8}$$

where $\varepsilon_1^e, \varepsilon_2^e$ and ε_6^e are components of the elastic strain tensor $\boldsymbol{\varepsilon}^e$, and the Voigt contracted notation (with $\varepsilon_6 = 2\varepsilon_{12}$) will be used in the rest of the paper.

In order to simplify the expressions of the above equation, we introduce the effective damage variables $d_i^{t,c}$ (i = 1, 2):

$$d_i^{t,c} = \begin{cases} d_i^t = d_i & \text{if } \sigma_i \geqslant 0 \\ d_i^c = d_i e^{-\langle-k_i\sigma_i\rangle^2} = d_i e^{-(k_i\sigma_i)^2} & \text{if } \sigma_i < 0 \end{cases} \tag{9}$$

where the superscripts t and c denote the tension and compression stress state respectively. It can be deduced that d_i^c in the equation progressively decreases as the compression stress σ_i increases for a given tensile damage variable d_i. By this means, the damage deactivation process of the material under compression loading is simulated.

Substituting Eq. (9) into Eq. (8), a simplified relation can be derived:

$$\boldsymbol{\varepsilon}^e = \boldsymbol{C}_d : \boldsymbol{\sigma} \tag{10}$$

where symbol (:) denotes inner product of two vectors or tensors; $\boldsymbol{C}_d$ represents the compliance matrix of the damaged material:

$$\boldsymbol{C}_d = \begin{bmatrix} \frac{1}{E_1^0\left(1-d_1^{t,c}\right)} & -\frac{\nu_{21}^0}{E_2^0} & 0 \\ -\frac{\nu_{12}^0}{E_1^0} & \frac{1}{E_2^0\left(1-d_2^{t,c}\right)} & 0 \\ 0 & 0 & \frac{1}{G_{12}^0(1-d_6)} \end{bmatrix} \tag{11}$$

in which $\nu_{21}^0/E_2^0 = \nu_{12}^0/E_1^0$, and ν_{21}^0 is another Poisson's ratio of the undamaged material.

Eq. (10) is equivalent to the following constitutive equation:

$$\boldsymbol{\sigma} = \boldsymbol{S}_d : \boldsymbol{\varepsilon}^e \tag{12}$$

where $\boldsymbol{S}_d$ is the stiffness matrix of the damaged material, and it is derived by inverting the compliance matrix $\boldsymbol{C}_d$ and similar to the forms presented in Refs.[23,24]:

$$\boldsymbol{S}_d = \frac{1}{t}\begin{bmatrix} E_1^0\left(1-d_1^{t,c}\right) & E_1^0\nu_{21}^0 d' & 0 \\ E_2^0\nu_{12}^0 d' & E_2^0\left(1-d_2^{t,c}\right) & 0 \\ 0 & 0 & tG_{12}^0(1-d_6) \end{bmatrix} \tag{13}$$

where $t = 1 - \nu_{12}^0\nu_{21}^0\left(1-d_1^{t,c}\right)\left(1-d_2^{t,c}\right), d' = \left(1-d_1^{t,c}\right)\left(1-d_2^{t,c}\right)$.

Further, we introduce the effective stress tensor $\tilde{\boldsymbol{\sigma}}$ based on the postulate of strain equivalent[23,24]:

$$\tilde{\boldsymbol{\sigma}} = \boldsymbol{S}_0 : \boldsymbol{\varepsilon}^e \tag{14}$$

where $\boldsymbol{S}_0$ is the stiffness matrix of the intact material.

By substituting Eq. (10) into Eq. (14), one can obtain the relationship between effective stress tensor $\tilde{\boldsymbol{\sigma}}$ and nominal stress tensor $\boldsymbol{\sigma}$:

$$\tilde{\boldsymbol{\sigma}} = \boldsymbol{S}_0 : \boldsymbol{C}_d : \boldsymbol{\sigma} \approx \boldsymbol{M} : \boldsymbol{\sigma} \tag{15}$$

where $\boldsymbol{M}$ denotes the damage operator with the expression:

$$\boldsymbol{M} = \begin{bmatrix} \frac{1}{1-d_1^{t,c}} & 0 & 0 \\ 0 & \frac{1}{1-d_2^{t,c}} & 0 \\ 0 & 0 & \frac{1}{1-d_6} \end{bmatrix} \tag{16}$$

In practice, the equation $\tilde{\boldsymbol{\sigma}} = \boldsymbol{M} : \boldsymbol{\sigma}$ is generally applied for the sake of simplicity,[23–25] which indeed brings in some calculation error when $d_1^{t,c} \neq d_2^{t,c}$. The same approximation will be adopted in the present analysis.

2.2. Damage evolution law

As observed in the test results,[3] the material damage starts evolution after the applied tensile and/or shear stresses exceeding the proportional limits, and its evolution rate is dependent on the loading state due to the coupling effects between the damage modes. Meanwhile, as the SiC matrix is insusceptible to compression loads, it is reasonable to assume that the in-plane compression stress will not cause any damage until the ultimate failure of the material.

According to the CDM theory,[20] the thermodynamic forces Y_i conjugated with the damage variables d_i (i = 1, 2, 6) are derived from the thermodynamic potential:

$$Y_i = \rho \frac{\partial \psi_e}{\partial d_i} \tag{17}$$

Substituting Eq. (2) into the above equation, we obtain

$$\begin{cases} Y_1 = \rho \dfrac{\partial \psi_e}{\partial d_1} = \dfrac{1}{2E_1^0}\left[\dfrac{\langle \sigma_1 \rangle^2}{(1-d_1)^2} + \dfrac{\partial \varphi(\langle -\sigma_1 \rangle, d_1)}{\partial d_1}\right] \\ Y_2 = \rho \dfrac{\partial \psi_e}{\partial d_2} = \dfrac{1}{2E_2^0}\left[\dfrac{\langle \sigma_2 \rangle^2}{(1-d_2)^2} + \dfrac{\partial \varphi(\langle -\sigma_2 \rangle, d_2)}{\partial d_2}\right] \\ Y_6 = \rho \dfrac{\partial \psi_e}{\partial d_6} = \dfrac{1}{2G_{12}^0}\dfrac{\sigma_6^2}{(1-d_6)^2} = \dfrac{\tilde{\sigma}_6^2}{2G_{12}^0} \end{cases} \tag{18}$$

The thermodynamic forces Y_1 and Y_2 in the above equation can be respectively divided into two parts:

$$Y_i = Y_i^+ + Y_i^- \quad (i = 1, 2) \tag{19}$$

where

$$\begin{cases} Y_i^+ = \dfrac{\langle \sigma_i \rangle^2}{2E_i^0(1-d_i)^2} = \dfrac{(H(\tilde{\sigma}_i)\tilde{\sigma}_i)^2}{2E_i^0} \geqslant 0 \\ Y_i^- = \dfrac{1}{2E_i^0}\dfrac{\partial \varphi(\langle -\sigma_i \rangle, d_i)}{\partial d_i} \\ \quad = \dfrac{1}{2E_i^0 k_i^2}\left[\dfrac{1}{1-d_i} - \dfrac{1}{e^{\langle -k_i\sigma_i \rangle^2} - d_i}\right] \geqslant 0 \end{cases} \tag{20}$$

where $\tilde{\sigma}_i$ is the component of the effective stress tensor $\tilde{\boldsymbol{\sigma}}$; and $H(x)$ is the Heaviside function:

$$H(x) = \begin{cases} 1 & \text{if } x \geqslant 0 \\ 0 & \text{if } x < 0 \end{cases} \tag{21}$$

Both $Y_{i=1,2}^+$ and $Y_{i=1,2}^-$ in Eq. (20) are non-negative and have the same unit (MPa), therefore $Y_{i=1,2,6} \geqslant 0$ in Eq. (18) for any in-plane stress state. Based on the damage characteristics of this material, we assume that the thermodynamic forces $Y_{i=1,2}^-$ associated with the compression stresses do not contribute to the damage evolution. In order to be thermodynamically consistent, the damage evolution laws must satisfy the following inequalities to fulfill the condition of positive dissipation:

$$\dot{d}_i \geqslant 0 \quad (i = 1, 2, 6) \tag{22}$$

Under these restrictions, the damage evolution laws are established[15,20]:

$$\begin{cases} d_1 = f_1(\langle Y_{1,\max}^* - Y_1^0 \rangle) \\ d_2 = f_1\left(\langle Y_{2,\max}^* - Y_1^0 \rangle\right) \\ d_6 = f_6\left(\langle Y_{6,\max}^* - Y_6^0 \rangle\right) \end{cases} \tag{23}$$

where f_1 and f_6 are the damage evolution functions; $Y_{i=1,2,6}^*$ are functions of the thermodynamic forces $Y_{i=1,2,6}$ representing the equations of damage surfaces, and the subscript "max" represents the maximum value of the variable recorded in the static loading history; Y_1^0 and Y_6^0 denote the initial damage thresholds.

The damage surfaces can be explicitly expressed as

$$\begin{cases} Y_1^* = Y_1^+ + g_{12}(Y_2^+)Y_2^+ \\ \quad +g_{16}(Y_6)Y_6 H(\tilde{\sigma}_1 + \tilde{\sigma}_2) \\ Y_2^* = Y_2^+ + g_{12}(Y_1^+)Y_1^+ \\ \quad +g_{16}(Y_6)Y_6 H(\tilde{\sigma}_1 + \tilde{\sigma}_2) \\ Y_6^* = Y_6 + g_{61}(Y_1^+)Y_1^+ \\ \quad +g_{61}(Y_2^+)Y_2^+ - g_c(\langle Z - Z^0 \rangle)\langle Z - Z^0 \rangle \end{cases} \tag{24}$$

where

$$Z = \frac{\langle -(\sigma_1 + \sigma_2) \rangle^2}{2E_1^0} \tag{25}$$

and g_{12}, g_{16} and g_{61} are monotonic increasing functions representing the damage coupling effects; the factor $H(\tilde{\sigma}_1 + \tilde{\sigma}_2)$ is introduced to eliminate the tensile damage that may be arisen by the shear stress through the coupling function g_{16} under combined compression and shear loads; Z is a new thermodynamic force[15] and g_c a related scalar function which are introduced to take into account the impediment effect of compression stress on the development of shear damage, and Z^0 is a positive material parameter representing the threshold of this impediment effect.

2.3. Evolution law of inelastic strain

The test results of the 2D plain-woven C/SiC composite[3] have revealed that the accumulation of inelastic strain relates to the evolution of microscopic damage such as matrix cracking, interface debonding and sliding or any combination of them. This inelastic strain is actually different from the plastic strain of ductile metals regarding their internal mechanisms; however, it can be macroscopically described within the framework of plasticity.[26,27] To date, a number of combined elasto-plastic damage models have been developed for the composites.[9–11,20,28,29]

In this study, we establish the evolution laws of inelastic strain for the 2D plain-woven C/SiC composite based on the plastic model proposed by Chen et al.,[24] in which an isotropic hardening law is adopted and the hysteresis effect is neglected. The "plastic" yield function f in terms of the effective stress tensor $\tilde{\boldsymbol{\sigma}}$ is defined as follows:

$$f(\tilde{\boldsymbol{\sigma}}, \bar{\varepsilon}^p) = \sqrt{H(\tilde{\sigma}_1)\tilde{\sigma}_1^2 + H(\tilde{\sigma}_2)\tilde{\sigma}_2^2 + m^2\tilde{\sigma}_6^2} - \kappa(\bar{\varepsilon}^p) = 0 \tag{26}$$

where m is a material constant proportional to the ratio of inelastic strain developed under the pure shear loading to that generated under the on-axis tensile loading at the same effective stress level; κ is the hardening function in terms of the equivalent inelastic strain $\bar{\varepsilon}^p$ and expressed as follows:

$$\kappa(\bar{\varepsilon}^p) = \bar{\sigma}(\bar{\varepsilon}^p) = R_0 + \beta(\bar{\varepsilon}^p)^\gamma \tag{27}$$

where R_0 is the initial threshold stress of inelastic strain; β and γ are material parameters; $\bar{\sigma}$ is the equivalent stress associated with the equivalent inelastic strain $\bar{\varepsilon}^{\mathrm{p}}$, and $\bar{\sigma}(\bar{\varepsilon}^{\mathrm{p}})$ represents the isotropic hardening law.

The associated plastic flow rule is applied for simplicity, therefore the inelastic strain rate $\dot{\boldsymbol{\varepsilon}}^{\mathrm{p}}$ can be expressed as[24]

$$\dot{\boldsymbol{\varepsilon}}^{\mathrm{p}} = \dot{\lambda}^{\mathrm{p}}\partial_{\tilde{\boldsymbol{\sigma}}}f \tag{28}$$

where $\partial_{\tilde{\boldsymbol{\sigma}}}f = \partial f/\partial\tilde{\boldsymbol{\sigma}}$; $\dot{\lambda}^{\mathrm{p}}$ is referred to as the non-negative plastic consistency parameter and assumed to obey the following Kuhn–Tucker loading/unloading conditions[30]:

$$\begin{cases} f \leqslant 0 \\ \dot{\lambda}^{\mathrm{p}} \geqslant 0 \\ \dot{\lambda}^{\mathrm{p}} f \equiv 0 \end{cases} \tag{29}$$

Substituting Eq. (26) into Eq. (28), the explicit form of inelastic strain rate is derived:

$$\begin{bmatrix} \dot{\varepsilon}_1^{\mathrm{p}} \\ \dot{\varepsilon}_2^{\mathrm{p}} \\ \dot{\varepsilon}_6^{\mathrm{p}} \end{bmatrix} = \dot{\lambda}^{\mathrm{p}}\partial_{\tilde{\boldsymbol{\sigma}}}f = \dot{\lambda}^{\mathrm{p}} \begin{bmatrix} H(\tilde{\sigma}_1)\tilde{\sigma}_1/\bar{\sigma} \\ H(\tilde{\sigma}_2)\tilde{\sigma}_2/\bar{\sigma} \\ m^2\tilde{\sigma}_6/\bar{\sigma} \end{bmatrix} \tag{30}$$

where $\bar{\sigma} = \sqrt{H(\tilde{\sigma}_1)\tilde{\sigma}_1^2 + H(\tilde{\sigma}_2)\tilde{\sigma}_2^2 + m^2\tilde{\sigma}_6^2}$.

The rate of plastic work per unit volume $\dot{W}^{\mathrm{p}}$ is given by

$$\dot{W}^{\mathrm{p}} = \tilde{\boldsymbol{\sigma}} : \dot{\boldsymbol{\varepsilon}}^{\mathrm{p}} = \bar{\sigma} \times \dot{\bar{\varepsilon}}^{\mathrm{p}} \tag{31}$$

Then, the equivalent plastic strain rate $\dot{\bar{\varepsilon}}^{\mathrm{p}}$ can be derived by substituting Eq. (30) into Eq. (31):

$$\dot{\bar{\varepsilon}}^{\mathrm{p}} = \dot{\lambda}^{\mathrm{p}} \tag{32}$$

From the above equations, one can deduce that the evolution law of inelastic strain can be obtained provided that the effective stress tensor $\tilde{\boldsymbol{\sigma}}$ and the plastic consistency parameter $\dot{\lambda}^{\mathrm{p}}$ are determined.

2.4. Failure criterion

In most cases, the composite structures are subjected to complex multiaxial loading conditions, and each stress component affects the ultimate strength and their interactions are not negligible. As a result, an appropriate failure criterion is needed for a prediction of the structural strength. Siron et al. separately used the $\partial\sigma/\partial\varepsilon \to 0$ condition and critical damage parameters $d_i^{\max}$ (i = 1, 2, 6) as failure criteria to predict the ultimate strengths of the 2.5D C/C composite plates subjected to off-axis tensile loads, and the latter criterion leads to a smaller discrepancy between the predicted and measured failure stresses[29]; Camus used the critical effective tensile stresses, a critical shear damage value and the maximum compression energy as failure flags for the 2D SiC/SiC composite in different stress states.[15] In recent years, the well-known average stress criterion, which is a nonlocal failure criterion and originally developed by Whitney and Nuismer for laminates,[31] was introduced to the fracture analysis of CMCs after some modifications.[11,32] In Refs.[11,32], it is assumed that the CMC structure reaches its ultimate strength when the average tensile stress over a circular or square area (defined by a characteristic length l_0) equals the unnotched tensile strength. This criterion has been proved to be applicable to CMC structures. However, a post-processing step is required to determine the final rupture state based on the FE simulation result. Multi-scale failure criteria developed for composite laminates, which are more physical mechanism based, can be found in Refs.[33,34].

In this study, for the sake of simplicity, we incorporate the local Tsai–Wu strength criterion[35] in the FE analysis. It takes the general form as follows:

$$f_{\mathrm{TW}} = F_i\sigma_i + F_{ij}\sigma_i\sigma_j = 1 \quad (i,j = 1,2,\ldots,6) \tag{33}$$

where F_i and F_{ij} are strength tensors of the second and fourth rank, respectively.

In the plane stress state, the above equation reduces to:

$$\begin{aligned} f_{\mathrm{TW}} = {} & F_1\sigma_1 + F_2\sigma_2 + F_{11}\sigma_1^2 \\ & + F_{22}\sigma_2^2 + F_{66}\sigma_6^2 + 2F_{12}\sigma_1\sigma_2 = 1 \end{aligned} \tag{34}$$

with:

$$\begin{cases} F_1 = \frac{1}{X_{\mathrm{t}}} - \frac{1}{X_{\mathrm{c}}} \\ F_2 = \frac{1}{Y_{\mathrm{t}}} - \frac{1}{Y_{\mathrm{c}}} \\ F_{11} = \frac{1}{X_{\mathrm{t}}X_{\mathrm{c}}} \\ F_{22} = \frac{1}{Y_{\mathrm{t}}Y_{\mathrm{c}}} \\ F_{66} = \frac{1}{S^2} \end{cases} \tag{35}$$

where X_{t} and X_{c} are the longitudinal tension and compression strengths respectively, Y_{t} and Y_{c} the transverse tension and compression strengths respectively, S the in-plane shear strength. Obviously, there are $X_{\mathrm{t}} = Y_{\mathrm{t}}$ and $X_{\mathrm{c}} = Y_{\mathrm{c}}$ for the 2D plain-woven C/SiC composite. F_{12} represents the interaction of two direct stress components (σ_1 and σ_2) and its determination is relatively difficult. A biaxial tension test is recommended[35] but it needs more complicated specimens and test instruments. The off-axis tension tests conducted in Ref.[3] do create combined stress states; however, the results can be merely referred because the stress fields are non-uniform near the ends of specimens due to the constraint effects. So we assume different values of F_{12} in the generalized tension and compression stress states:

$$F_{12} = \begin{cases} F_{12}^{\mathrm{t}} & (\sigma_1 + \sigma_2) \geqslant 0 \\ F_{12}^{\mathrm{c}} & (\sigma_1 + \sigma_2) < 0 \end{cases} \tag{36}$$

where F_{12}^{t} and F_{12}^{c} are determined through inverse identification.

3. Numerical implementation

The proposed non-linear constitutive model is implemented into the commercial FE software ABAQUS via a user-defined material subroutine (UMAT) programmed in FORTRAN language, in which the numerical integration algorithms for updating the solution-dependent state variables (SDVs) and the nominal stress tensor $\boldsymbol{\sigma}$ must be defined, and the material Jacobian matrix (i.e. the consistent tangent stiffness matrix) should be provided in order to achieve a rapid convergence rate.[36]

3.1. Integration algorithm

The material model takes into account the evolution laws of damage and inelastic strain, hence its FE implementation requires numerical integrations over discrete time increments. In the incremental approach driven by strain, the loading history is discretized into a sequence of time steps $[t_n, t_{n+1}]$, $n \in \{0, 1, 2, 3, \ldots\}$, where each step is referred to as the $(n + 1)$th increment.[24] For a given strain increment $\Delta\boldsymbol{\varepsilon}_{n+1}$, the variables $\varepsilon_i^{\mathrm{p}}, \bar{\sigma}, \bar{\varepsilon}^{\mathrm{p}}, Y_i^+, Y_{i,\max}^*, \tilde{\sigma}_i,$ $d_i,$ $d_{1,2}^{\mathrm{t,c}},$ $f_{\mathrm{TW,max}}$ (i = 1, 2, 6) in the material model are defined as SDVs and they are passed in known variables at the beginning of the $(n+1)$th increment, where $f_{\mathrm{TW,max}}$ records the failure state of the element.

The backward Euler implicit integration procedure is adopted,[24,37] and the integration algorithm based on the material model is formulated as follows:

$$\begin{cases} \boldsymbol{\varepsilon}_{n+1} = \boldsymbol{\varepsilon}_n + \Delta\boldsymbol{\varepsilon}_{n+1} \\ \boldsymbol{\varepsilon}^{\mathrm{p}}_{n+1} = \boldsymbol{\varepsilon}^{\mathrm{p}}_n + \Delta\lambda^{\mathrm{p}}_{n+1}\partial_{\tilde{\boldsymbol{\sigma}}_{n+1}} f_{n+1} \\ \bar{\varepsilon}^{\mathrm{p}}_{n+1} = \bar{\varepsilon}^{\mathrm{p}}_n + \Delta\lambda^{\mathrm{p}}_{n+1} \\ \tilde{\boldsymbol{\sigma}}_{n+1} = \boldsymbol{S}_0 : (\boldsymbol{\varepsilon}_{n+1} - \boldsymbol{\varepsilon}^{\mathrm{p}}_{n+1}) \\ f_{n+1}(\tilde{\boldsymbol{\sigma}}_{n+1}, \bar{\varepsilon}^{\mathrm{p}}_{n+1}) \leqslant 0 \\ \boldsymbol{\sigma}_{n+1} = \boldsymbol{S}_{\mathrm{d},n+1} : (\boldsymbol{\varepsilon}_{n+1} - \boldsymbol{\varepsilon}^{\mathrm{p}}_{n+1}) \end{cases} \tag{37}$$

where $\boldsymbol{\varepsilon}_n$, $\Delta\boldsymbol{\varepsilon}_{n+1}$ and $\boldsymbol{\varepsilon}_{n+1}$ are known quantities; $\Delta\lambda^{\mathrm{p}}_{n+1}, \bar{\varepsilon}^{\mathrm{p}}_{n+1}, \boldsymbol{\varepsilon}^{\mathrm{p}}_{n+1}, \tilde{\boldsymbol{\sigma}}_{n+1}$, $\boldsymbol{S}_{\mathrm{d},n+1}$ and $\boldsymbol{\sigma}_{n+1}$ are unknown SDVs to be solved.

Firstly, the strain increment $\Delta\boldsymbol{\varepsilon}_{n+1}$ is assumed to be an elastic strain increment $\Delta\boldsymbol{\varepsilon}^{\mathrm{e}}_{n+1}$, and then we can obtain a trial effective stress tensor $\tilde{\boldsymbol{\sigma}}^{\mathrm{trial}}_{n+1}$ and substitute it into the yield function f_{n+1} in Eq. (37). If the inequality $f_{n+1} \leqslant 0$ is satisfied, then the material stays in the elastic domain or elastic unloading–reloading state; otherwise, the plastic loading happens and there is an inelastic strain increment $\Delta\boldsymbol{\varepsilon}^{\mathrm{p}}_{n+1}$. From Eq. (37) one can find that the variables $\boldsymbol{\varepsilon}^{\mathrm{p}}_{n+1}$, $\bar{\varepsilon}^{\mathrm{p}}_{n+1}$, $\tilde{\boldsymbol{\sigma}}_{n+1}$ are related to the increment of the plastic consistency parameter $\Delta\lambda^{\mathrm{p}}_{n+1}$, therefore the problem can be reduced to calculate the value of $\Delta\lambda^{\mathrm{p}}_{n+1}$ to make $f_{n+1} = 0$, i.e.:

$$f\left(\tilde{\boldsymbol{\sigma}}\left(\Delta\lambda^{\mathrm{p}}_{n+1}\right), \bar{\varepsilon}^{\mathrm{p}}\left(\Delta\lambda^{\mathrm{p}}_{n+1}\right)\right) = 0 \tag{38}$$

The Newton–Raphson iteration method is used to solve the above function. The initial conditions are set as: $\Delta\lambda^{\mathrm{p},(0)}_{n+1} = 0, \tilde{\boldsymbol{\sigma}}^{(0)}_{n+1} = \tilde{\boldsymbol{\sigma}}^{\mathrm{trial}}_{n+1}, \boldsymbol{\varepsilon}^{\mathrm{p},(0)}_{n+1} = \boldsymbol{\varepsilon}^{\mathrm{p}}_n, \bar{\varepsilon}^{\mathrm{p},(0)}_{n+1} = \bar{\varepsilon}^{\mathrm{p}}_n$, and then the following iterations are performed:

$$\Delta\lambda^{\mathrm{p},(k+1)}_{n+1} = \Delta\lambda^{\mathrm{p},(k)}_{n+1} - \frac{f\left(\Delta\lambda^{\mathrm{p},(k)}_{n+1}\right)}{f'\left(\Delta\lambda^{\mathrm{p},(k)}_{n+1}\right)} \tag{39}$$

until $f\left(\tilde{\boldsymbol{\sigma}}\left(\Delta\lambda^{\mathrm{p},(k+1)}_{n+1}\right), \bar{\varepsilon}^{\mathrm{p}}\left(\Delta\lambda^{\mathrm{p},(k+1)}_{n+1}\right)\right) \leqslant \mathrm{TOL}$, where the superscript $k \in \{0, 1, 2, 3, \ldots\}$, and TOL is the error tolerance and set to 1.0×10^{-8} in the present work. The calculated results $\Delta\lambda^{\mathrm{p},(k+1)}_{n+1}, \boldsymbol{\varepsilon}^{\mathrm{p},(k+1)}_{n+1}, \bar{\varepsilon}^{\mathrm{p},(k+1)}_{n+1}$ and $\tilde{\boldsymbol{\sigma}}^{(k+1)}_{n+1}$ are stored as updated SDVs.

Since the effective stress tensor $\tilde{\boldsymbol{\sigma}}_{n+1}$ has been renewed, the thermodynamic forces $Y^{+}_{1,n+1}, Y^{+}_{2,n+1}$, Y_6 and their functions $Y^{*}_{i,n+1}$, damage variables $d_{i,n+1}$ $(i = 1, 2, 6)$ and the damaged stiffness matrix $\boldsymbol{S}_{\mathrm{d},n+1}$ can be successively calculated. Finally, the nominal stress $\boldsymbol{\sigma}_{n+1}$ can be determined.

If the stresses on the integration points are instantly removed once the Tsai–Wu failure criterion is fulfilled in local elements, this would probably lead to a convergence problem. Then the FE analysis might abort and the structure's load bearing capacity would be underestimated. In order to avoid this problem, we assume $\Delta\boldsymbol{\varepsilon}^{\mathrm{p}}_{n+1} = \Delta\boldsymbol{\varepsilon}_{n+1}$ for the failed elements, then according to Eq. (37) the effective stress tensor will remain unchanged (i.e. $\tilde{\boldsymbol{\sigma}}_{n+1} = \tilde{\boldsymbol{\sigma}}_n$), and the damage variables will also keep constant. In this way, the failed material is supposed to show the ideal plastic behavior in the post-failure regime. Indeed, this assumption would lead to more dissipated energy and overestimate the structural strength; however, it will effectively alleviate the convergence problem. So it can be seen as a compromise between the computational difficulty and accuracy.

3.2. Jacobian matrix

The consistency condition requires that the stress points stay on the yield surface during the loading process. For the isotropic hardening material, the following equations must be satisfied:

$$\begin{cases} f(\tilde{\sigma}_{ij}, \bar{\varepsilon}^{\mathrm{p}}) = 0 \\ f(\tilde{\sigma}_{ij} + \mathrm{d}\tilde{\sigma}_{ij}, \bar{\varepsilon}^{\mathrm{p}} + \mathrm{d}\bar{\varepsilon}^{\mathrm{p}}) = 0 \end{cases} \tag{40}$$

Thus, we can write

$$\mathrm{d}f = 0 \Rightarrow \frac{\partial f}{\partial \tilde{\sigma}_{ij}}\mathrm{d}\tilde{\sigma}_{ij} + \frac{\partial f}{\partial \kappa}\frac{\partial \kappa}{\partial \bar{\varepsilon}^{\mathrm{p}}}\mathrm{d}\bar{\varepsilon}^{\mathrm{p}} = 0 \tag{41}$$

Substituting Eqs. (14), (26), (27), (28) and (32) into the above function, we obtain

$$\mathrm{d}\lambda^{\mathrm{p}} = \frac{\partial_{\tilde{\boldsymbol{\sigma}}} f : \boldsymbol{S}_0 : \mathrm{d}\boldsymbol{\varepsilon}}{H_{\mathrm{p}} + \partial_{\tilde{\boldsymbol{\sigma}}} f : \boldsymbol{S}_0 : \partial_{\tilde{\boldsymbol{\sigma}}} f} \tag{42}$$

where H_{p} is the plastic modulus and defined as follows:

$$H_{\mathrm{p}} = \frac{\partial \kappa}{\partial \bar{\varepsilon}^{\mathrm{p}}} = \beta\gamma(\bar{\varepsilon}^{\mathrm{p}})^{\gamma-1} \tag{43}$$

Besides, Eq. (12) has the following incremental form:

$$\mathrm{d}\boldsymbol{\sigma} = \boldsymbol{S}_{\mathrm{d}} : \mathrm{d}\boldsymbol{\varepsilon}^{\mathrm{e}} = \boldsymbol{S}_{\mathrm{d}} : (\mathrm{d}\boldsymbol{\varepsilon} - \mathrm{d}\lambda^{\mathrm{p}}\partial_{\tilde{\boldsymbol{\sigma}}} f) \tag{44}$$

Substituting Eq. (42) into the above equation, an incremental form of the stress/strain relationship is derived:

$$\mathrm{d}\boldsymbol{\sigma} = \boldsymbol{S}_{\mathrm{dp}} : \mathrm{d}\boldsymbol{\varepsilon} \tag{45}$$

where

$$\boldsymbol{S}_{\mathrm{dp}} = \boldsymbol{S}_{\mathrm{d}} - \frac{(\boldsymbol{S}_{\mathrm{d}} : \partial_{\tilde{\boldsymbol{\sigma}}} f) \otimes (\boldsymbol{S}_0 : \partial_{\tilde{\boldsymbol{\sigma}}} f)}{H_{\mathrm{p}} + \partial_{\tilde{\boldsymbol{\sigma}}} f : \boldsymbol{S}_0 : \partial_{\tilde{\boldsymbol{\sigma}}} f} \tag{46}$$

where $\boldsymbol{S}_{\mathrm{dp}}$ is the consistent tangent matrix of the material model, and $(\otimes)$ denotes a tensor product.

The numerator $(\boldsymbol{S}_{\mathrm{d}} : \partial_{\tilde{\boldsymbol{\sigma}}} f) \otimes (\boldsymbol{S}_0 : \partial_{\tilde{\boldsymbol{\sigma}}} f)$ in the above equation is asymmetric for most loading conditions. In order to accelerate the convergence rate, the matrix $\boldsymbol{S}_{\mathrm{dp}}$ is symmetrized to be the following Jacobian matrix $\boldsymbol{J}_{\mathrm{dp}}$:

$$\boldsymbol{J}_{\mathrm{dp}} = \boldsymbol{S}_{\mathrm{d}} - \frac{[(\boldsymbol{S}_{\mathrm{d}} : \partial_{\tilde{\boldsymbol{\sigma}}} f) \otimes (\boldsymbol{S}_0 : \partial_{\tilde{\boldsymbol{\sigma}}} f) + (\boldsymbol{S}_0 : \partial_{\tilde{\boldsymbol{\sigma}}} f) \otimes (\boldsymbol{S}_{\mathrm{d}} : \partial_{\tilde{\boldsymbol{\sigma}}} f)]}{2(H_{\mathrm{p}} + \partial_{\tilde{\boldsymbol{\sigma}}} f : \boldsymbol{S}_0 : \partial_{\tilde{\boldsymbol{\sigma}}} f)} \tag{47}$$

As the variables in the above equation have been solved through the integration algorithm, the matrix $\boldsymbol{J}_{\mathrm{dp}}$ can be explicitly determined.

In the post-failure regime, the consistent tangent matrix $\boldsymbol{S}_{\mathrm{dp}}$ in Eq. (45) is equal to 0, and the corresponding Jacobian matrix is assumed to be $\boldsymbol{J}^{\mathrm{f}}_{\mathrm{dp}} = 0.01\boldsymbol{S}_0$ for numerical reasons, where the superscript f denotes the failure state.

4. Model parameters

The parameters in the model were determined mainly based on the test results in the first part of the paper.[3] Table 1 listed the elastic constants and some strength parameters obtained through the on-axis tension, compression and pure shear tests,

Table 1 Elastic constants and strength parameters of the 2D plain-woven C/SiC composite.

Property	Symbol	Value
Tensile/compressive modulus (GPa)	$E^0_{1,2}$	146.51
Poisson's ratio	v^0_{12}	0.067
Shear modulus (GPa)	G^0_{12}	37.82
Tensile strength (MPa)	X_t, Y_t	265.28
Compressive strength (MPa)	X_c, Y_c	338.94
Shear strength (MPa)	S	143.43
Strength parameter (MPa^{-2})	F^t_{12}	2.0×10^{-5}
Strength parameter (MPa^{-2})	F^c_{12}	2.5×10^{-6}

and the other strength parameters F^t_{12} and F^c_{12} in Eq. (36) determined by means of inverse identification.

As $Y^*_1 = Y^+_1$ and $Y^*_6 = Y_6$ in the on-axis tension and pure shear tests (see Eq. (24)), the incremental loading–unloading test method enables one to obtain the evolution laws of damage variables d_1 and d_6 as a function of Y^*_1 and Y^*_6 respectively, and the discrete data shown in Fig. 1 was fitted by the following logistic functions:

$$d_i = f_i(\langle Y^*_{i,\max} - Y^0_i \rangle) = a_i - \frac{a_i}{[1 + (\langle Y^*_{i,\max} - Y^0_i \rangle / b_i)^{c_i}]} \quad (i = 1, 6) \tag{48}$$

where a_i, b_i and c_i are shape parameters of the logistic function, in which a_i can be regarded as the ultimate values of the related damage variable d_i, and b_i is the normalizing thermodynamic force. The parameters identified are $Y^0_1 = 1.35 \times 10^{-3}$ MPa, $a_1 = 0.87$, $b_1 = 0.47$ MPa, $c_1 = 0.57$; $Y^0_6 = 2.17 \times 10^{-3}$ MPa, $a_6 = 0.86$, $b_6 = 0.39$ MPa, $c_6 = 0.63$.

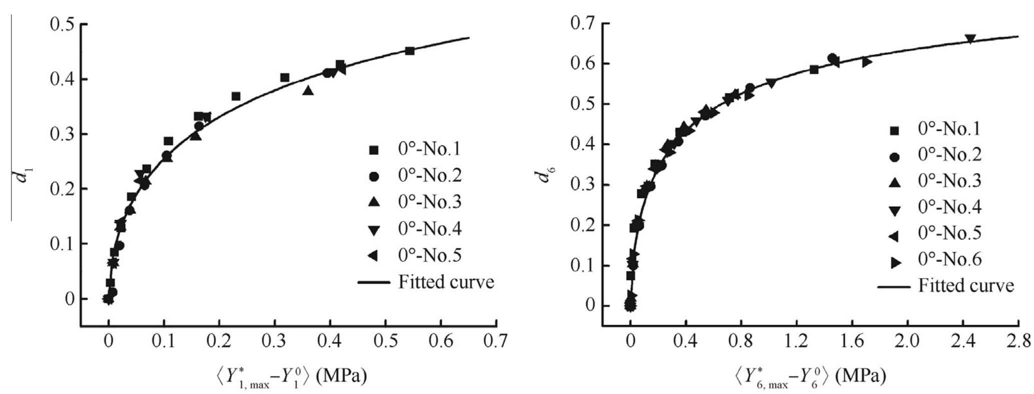
The damage coupling functions g_{12}, g_{16} and g_{61} in Eq. (24) were determined based on the experiment data obtained from the 15°, 30° and 45° off-axis incremental loading–unloading tensile tests, in which there are: $Y_1 = Y^+_1$, $Y_2 = Y^+_2$ and $Z = 0$, and the strains are measured close to the center of the specimen's gauge section, where the non-uniformity of the stress field caused by the end constraints and material asymmetry is moderately reduced.

Firstly, we notice that $Y_2 << Y_1$, Y_6 during the 15° off-axis tension test, therefore we assume $Y^*_1 = Y_1 + g_{16}(Y_6)Y_6$ as an approximation of Eq. (24), and then the coupling function $g_{16}(Y_6)$ can be separated because the variables d_1, Y_1 and Y_6 can be experimentally determined and Y^*_1 can be deduced from the identified damage evolution law f_1 in Eq. (48). In order to reduce the dispersion of experiment data, the whole term $g_{ij}(Y_j)Y_j$ is analyzed as a function of Y_j. Fig. 2(a) shows the relationship between $g_{16}(Y_6)Y_6$ and Y_6, in which a linear function is used to fit the discrete data:

$$g_{16}(Y_6)Y_6 = a_{16}Y_6 \tag{49}$$

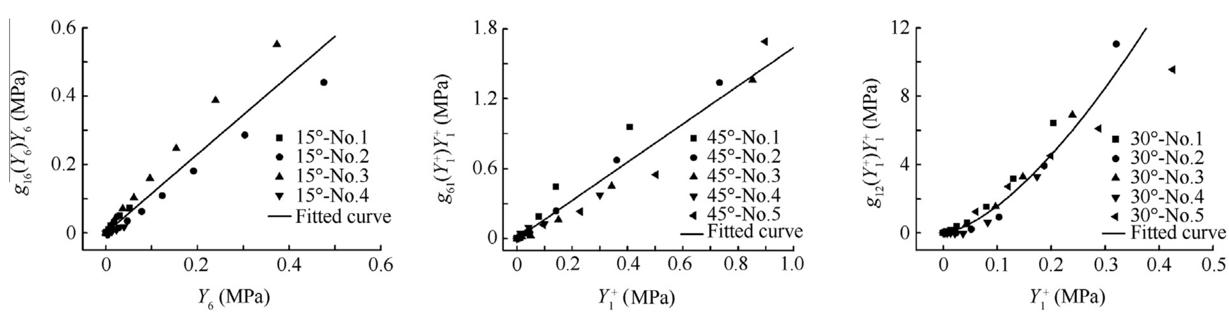
where a_{16} is a proportional constant and identified as $a_{16} = 1.15$.

Secondly, as $Y^+_1 = Y^+_2$ for the 45° off-axis tensile test results, the variation of $g_{61}(Y^+_1)Y^+_1$ as a function of Y^+_1 in Eq. (24) can be identified since the variables d_6, Y_6 and Y^+_1 can be determined and Y^*_6 can be deduced from the identified



(a) On-axis tension (b) Pure shear

Fig. 1 Damage evolution laws obtained from test results and fitted curves.



(a) $g_{16}(Y_6)Y_6$ (MPa) (b) $g_{61}(Y^+_1)Y^+_1$ (MPa) (c) $g_{12}(Y^+_1)Y^+_1$ (MPa)

Fig. 2 Experimental data and fitted curves of coupling functions.

damage evolution law f_6 in Eq. (48). A linear function was also applied to fit the discrete data (shown in Fig. 2(b)):

$$\begin{cases} g_{61}(Y_1^+)Y_1^+ = a_{61}Y_1^+ \\ g_{61}(Y_2^+)Y_2^+ = a_{61}Y_2^+ \end{cases} \tag{50}$$

where a_{61} is a proportional constant and identified as $a_{61} = 16.39$.

Thirdly, since the expression of $g_{16}(Y_6)Y_6$ has been known, the variation of $g_{12}(Y_1^+)Y_1^+$ as a function of Y_1^+ can be determined based on the 30° off-axis tensile test results through a similar procedure, and a power function is applied to fit the discrete data (shown in Fig. 2(c)):

$$g_{12}(Y_1^+)Y_1^+ = a_{12}(Y_1^+)^{b_{12}} \tag{51}$$

where a_{12} and b_{12} are shape parameters of the power function and identified as $a_{12} = 53.48$, $b_{12} = 1.53$.

Similarly, $g_{12}(Y_2^+)Y_2^+$ has the following expression:

$$g_{12}(Y_2^+)Y_2^+ = a_{12}(Y_2^+)^{b_{12}} \tag{52}$$

The impediment effect of compression stress on the evolution of shear damage is represented by the function g_c in Eq. (24), and it can be identified from the results of the off-axis compression tests as there are $Y_1^+ = 0$ and $Y_2^+ = 0$. The function g_c was determined by fitting the related 30° and 45° off-axis compression test results as shown in Fig. 3:

$$g_c(\langle Z - Z^0 \rangle) = a_c\left(1 - e^{-\langle Z - Z^0 \rangle / b_c}\right) \tag{53}$$

where a_c and b_c are shape parameters of the exponential function; Z^0 is determined from the initiation stress of the shear damage. They are identified as: $a_c = 0.81$, $b_c = 0.041$ MPa and $Z^0 = 0.017$ MPa.

The parameters in the evolution law of inelastic strain were identified based on the incremental loading–unloading tensile and shear test results. From Eqs. (26), (27) and (32), we can deduce that there are equations $\bar{\sigma} = \tilde{\sigma}_1$ and $\bar{\varepsilon}^p = \lambda^p = \varepsilon_1^p$ under the on-axis tensile loading, and hence the hardening function $\kappa(\bar{\varepsilon}^p)$ in Eq. (27) can be determined based on the uniaxial tensile test results. Comparatively, there are $\bar{\sigma} = m\tilde{\sigma}_6$ and $\bar{\varepsilon}^p = \lambda^p = m^{-1}\varepsilon_6^p$ under the pure shear loading, then the material constant m in the yield function can be inversely identified through the comparison between the related tensile and shear test results. The experimental data and a fitted curve are shown

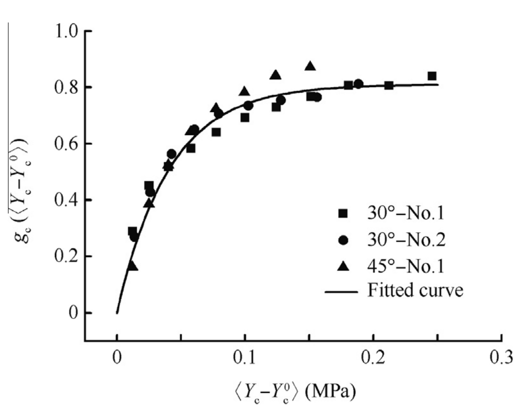


Fig. 3 Experimental data and a fitted curve of impediment function g_c.

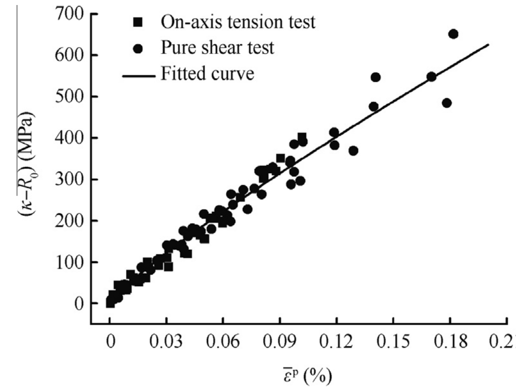


Fig. 4 Experimental results and a fitted curve of $(\kappa - R_0)$ versus equivalent plastic strain $\bar{\varepsilon}^p$.

in Fig. 4, and the identified parameters are: $R_0 = 18.79$ MPa, $m = 1.54$, $\beta = 1.31 \times 10^5$ MPa, $\gamma = 0.86$.

The damage deactivation parameters α_{11}, α_{12} in Eq. (5) were determined by inverse identification $\alpha_{11} = 0.195$, $\alpha_{12} = 1.365$, and the thermal residual stress σ_r in Eq. (4) was set to 130 MPa.[38]

5. Experimental validation and discussion

The material model was implemented as a UMAT subroutine and linked to ABAQUS. For the validation purpose, the axial stress/strain curves of unnotched on-axis and off-axis specimens[3] subjected to tension, compression and tension–compression loadings were simulated and compared with the experimental results. Further, as the C/SiC composites generally show moderate notch insensitivity in comparison with the monolithic ceramics,[39] which is attributable to the stress redistribution mechanisms including matrix microcracking[40] and the inelastic straining prior to the failure[41], we additionally simulate the tensile response of a 45° off-axis plate with two centrally located V-notches, and a corresponding experiment was also performed for comparison.

In each FE model, a local coordinate system was built to assign the off-axis angle (θ) of the reinforcing fabrics, and continuum shell elements with reduced integration and hourglass control (designated CPS4R in ABAQUS) were used to mesh the model. Except for the monotonic loading cases, a sequence of static analysis steps was created to simulate the incremental loading–unloading test processes.

5.1. On-axis and off-axis plates

Straight plates with the same dimensions of tension and compression specimens as illustrated in Ref.[3] were modeled and subjected to axial tension, compression and incremental tension–compression loads with different off-axis angles (θ). For each plate, the transverse displacements at both ends are fixed ($u_y = 0$) to keep consistent with the constraint condition in the experiment, and the element size in the longitudinal and transverse direction is set to 1.25 mm and 0.5 mm respectively.

Firstly, the global stress/strain relationships of the plates subjected to axial tension and compression loads were extracted from the simulation results and then compared with

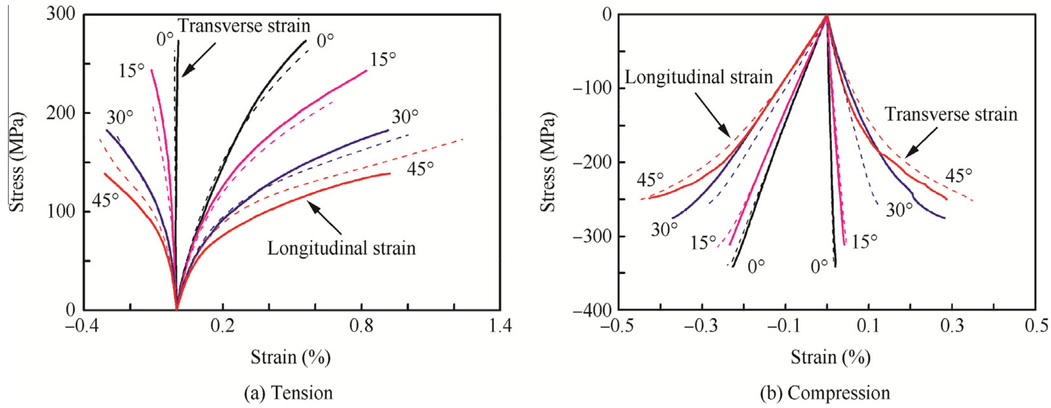


Fig. 5 Experimental (solid lines) and FE simulated (dashed lines) global stress/strain relationships of on- and off-axis plates under different loadings.

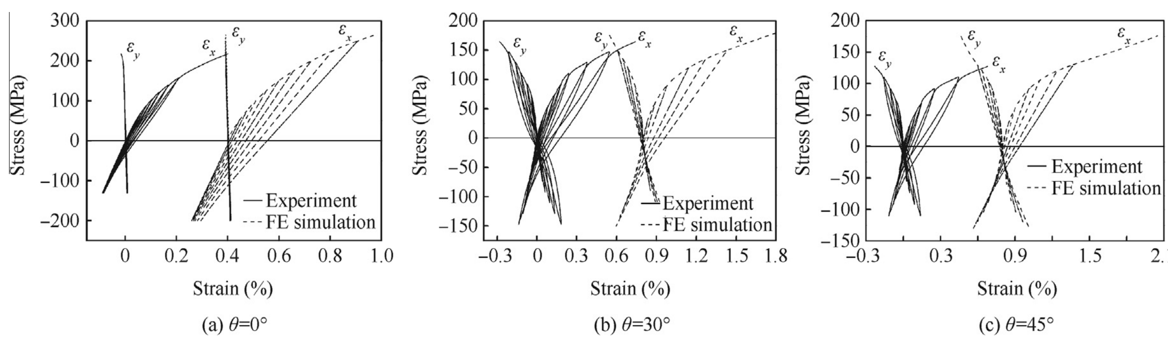


Fig. 6 Typical experimental and FE simulated tension–compression stress/strain curves of plates with different off-axis angles.

the typical experiment curves (Fig. 5). Since the material model has taken into account the damage evolution and accumulation of the inelastic strain in the material coordinate system, the global non-linear tensile responses of the on- and off-axis specimens can be predicted by the simulation results (see Fig. 5(a)). In comparison with the experiment results, the predicted strength of the 15° off-axis tension plate is underestimated to some extent, which may be attributed to the relatively large dispersion of the experiment data as obtained in Ref.[3]. Besides, the simulated stress/strain curves of the 45° off-axis tension plate are basically higher than the experiment results. This is mainly caused by the factor that the isotropic damage modes such as the matrix microcracks oriented by the axial load are not considered in the present model. Compression behavior of the plates, which is rather different from the tension test results, can be moderately predicted by the material model (see Fig. 5(b)) because the evolution of damage and inelastic strain is partially or fully impeded by the compression stress components.

Further, the mechanical responses of on- and off-axis plates subjected to the tension–compression loadings were simulated by creating a sequence of static tension and compression loading steps in the FE software. The global stress/strain curves obtained from the experiment and FE simulation results are presented in Fig. 6, where the simulated global mechanical responses are reflections of the damage or damage deactivation behaviors in the material coordinate system. During the axial tension loading, the progressive stiffness degradation and accumulation of inelastic strain can be observed in the simulated stress/strain curves. After the load reversal, the experiment curves reveal that damaged stiffness of the on-axis, 30° and 45° off-axis plates gradually recovers as the compression stress increases. These damage deactivation behavior can be moderately predicted by the present model; meanwhile, the continuity of the stress/strain curves is retained.

5.2. 45° off-axis plate with V-notches

The tensile response of a 45° off-axis plate with two centrally located V-notches (see Fig. 7) was simulated and then compared with the experiment result for further validation of the material model. In the experiment, strain gages designated SG1 and SG2 were separately affixed to the specimen to measure the tensile strains at the edge of the notch and the midpoint on the minimum cross section. The grid size of each strain gage is 3.0 mm by 4.0 mm, which is respectively equal to the total length of 40 and 16 elements along the x and y axis in the FE model. Due to the symmetric geometry and loading conditions as well as the material symmetries, 1/4th of the plate with locally refined meshes (see Fig. 7) was modeled to save computation resources.

Three strain components ε_1, ε_2 and ε_6 were extracted from the integration points of the elements covered by the strain gages SG1 and SG2. Then, the axial strain ε_x was calculated using the transformation formula $\varepsilon_x = (\varepsilon_1 + \varepsilon_2 - \varepsilon_6)/2$, and hence the curves of the nominal tensile stress σ_x in the net-section versus the strain ε_x at these two locations were derived from the simulation result and then compared with the experiment data, which were shown in Fig. 8. It can be observed that the simulated curves accord well with the experiment data at

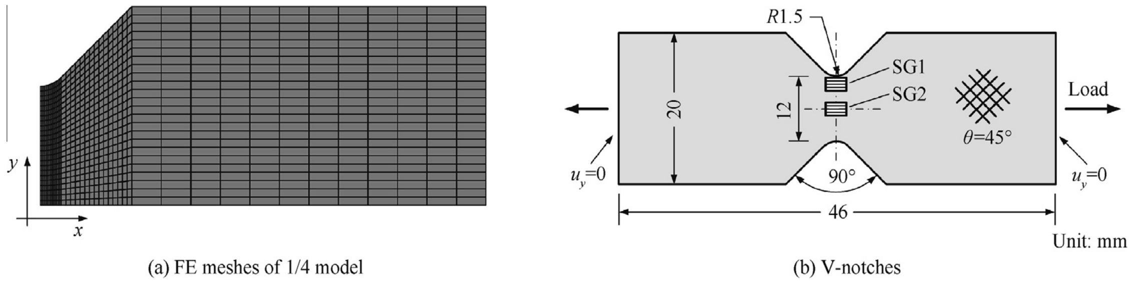


Fig. 7 FE meshes of 1/4 model and geometry and boundary conditions of 45° off-axis plate with V-notches.

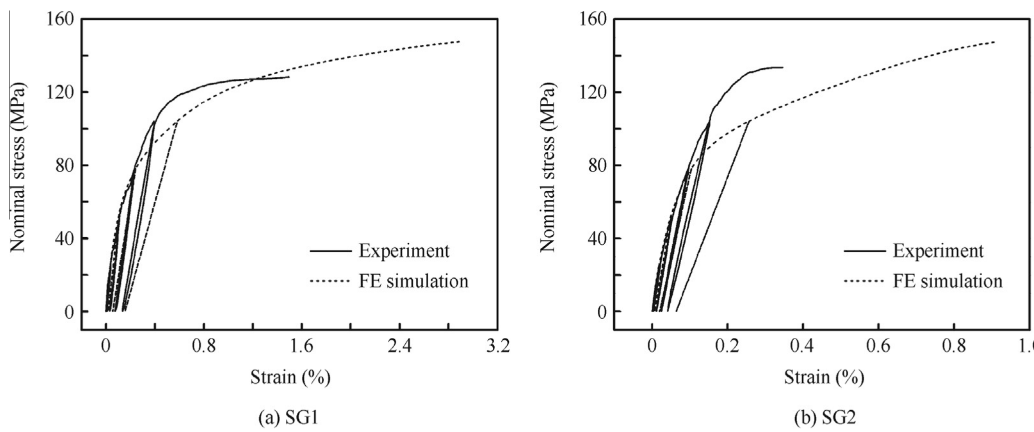


Fig. 8 Typical experiment data and simulated results of nominal tensile stress versus longitudinal strains at locations covered by SG1 and SG2 in the minimum cross section.

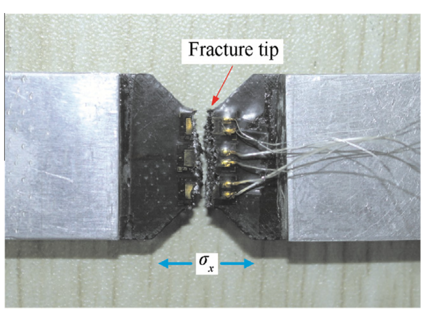


Fig. 9 Fractured 45° off-axis specimen with V-notches.

the stress level of around 80 MPa, after that they show higher stiffness degradation rates and accumulation rates of inelastic strain. Similar to the discrepancy between the predicted and experimental strength of the unnotched 45° off-axis tension plate (see Fig. 5(a)), the material model a little overestimates the nominal tensile strength of the V-notched 45° off-axis plate.

Apparent strain concentration near the edge of the notch can also be observed in Fig. 8, and the strain gage SG1 failed before the ultimate fracture of the specimen. Besides, a plateau appears in the experimental stress/strain curves, which is especially observable in Fig. 8(a). This phenomenon may be caused by large inelastic straining and delamination of the material, because the axial tensile stress/strain curve of the unnotched 45° off-axis plate in Fig. 5(a) shows a large fracture strain and it nearly reaches a similar plateau; meanwhile, the delamination can be detected in the fractured V-notched specimen (see Fig. 9). Probably due to these energy dissipation mechanisms, the nominal tensile strength in the minimum cross section (133.58 MPa) is not evidently affected by the stress concentration near the edge of the notch, considering that the average tensile strength of the unnotched 45° off-axis specimens is 125.08 ± 11.81 MPa.

A typical photo of the fractured specimen is presented in Fig. 9 and the deformed configuration of the simulated plate with stress and SDV contours following the ultimate load are presented in Fig. 10, where S_{11}, S_{22}, S_{12} and SDV20 correspond to σ_1, σ_2, σ_6 and $f_{\mathrm{TW,max}}$ respectively. Fig. 9 shows that the fracture plane propagates approximately along the minimum cross section of the specimen. This failure process of the plate can be predicted by the failure contour of SDV20 in Fig. 10(d). Besides, as large inelastic straining occurred at the failed elements as supposed in the material model, the contours of stress components (see Fig. 10(a)–(c)), especially σ_1 and σ_2, show that stress concentration in and near the failed area is not obvious due to the stress redistribution. These results also prove that inelastic straining would reduce the notch sensitivity of the material.

6. Conclusions

(1) In this paper, we have proposed a non-linear macroscopic constitutive model for the 2D plain-woven C/SiC composite. The evolution laws of damage and inelastic strain have been established based on the CDM and general plasticity theories, respectively. A

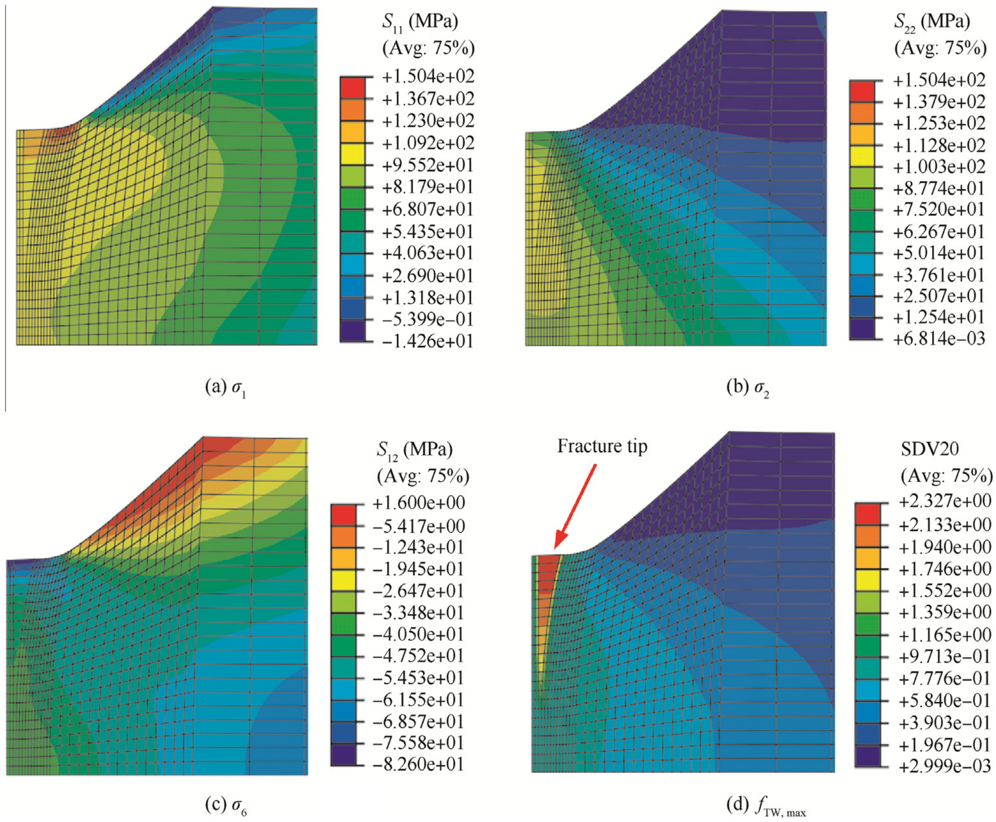


Fig. 10 Deformed configuration (amplified 20 times) of simulated plate with contours of stress components and failure state following ultimate load.

new thermodynamic potential function was developed, which enables the constitutive equations to take the damage, unilateral and damage deactivation behavior into consideration. Different damage deactivation rates in uniaxial and biaxial compression state were also considered. In order to achieve the numerical application of the proposed material model, the material Jacobian matrix was explicitly defined and the backward Euler implicit integration procedure was adopted.

(2) The main advantage of this material model is that its formulation is rather simple, which makes it convenient to be programmed and then implemented into the commercial FE software. Besides, most of the model parameters can be determined based on the on- and off-axis test results. The mechanical behavior of on- and off-axis plates subjected to different in-plane loadings and a 45° off-axis tension plate with V-notches was simulated in the ABAQUS software and then compared with the experiment results. The numerical results show that the non-linear mechanical response resulting from the development of damage and inelastic strain can be moderately captured by the model; moreover, the continuity of the stress/strain relationship during the damage deactivation process is comprehensibly retained. In addition, Tsai-Wu failure criterion seems to be adoptable in the strength analysis. The model, after some adjustments, can be applied to the unidirectional fiber-reinforced C/SiC composite laminates which also show elasto-plastic-damage behavior.

(3) Some damage mechanisms of the material such as the isotropic microcracking and the deactivation process of shear damage under compression loadings are not considered in the present model. This leads to some discrepancies between the experiment and numerical results. Besides, the availability of the model is confined to in-plane loading conditions. Future work is needed to overcome these inadequacies.

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Appendix A.

The positive definition of the complementary elastic strain energy density $\rho\psi_e$ in Eq. (2) requires that the inequation $\rho\psi_e > 0$ should be fulfilled for all nonzero stress vectors $\boldsymbol{\sigma}$ $(=[\sigma_1,\ \sigma_2,\ \sigma_6]^T)$. Considering that the stress states can be divided into four types according to the signs of the normal stress components, we can discuss the positive definiteness of $\rho\psi_e$ as follows:

1. If $\sigma_1 > 0$, $\sigma_2 > 0$, the corresponding expression of $\rho\psi_e$ in Eq. (2) reduces to

$$\rho\psi_{\mathrm{e}}=\frac{1}{2}\left[\frac{\sigma_1^2}{E_1^0(1-d_1)}-\frac{2v_{12}^0\sigma_1\sigma_2}{E_1^0}+\frac{\sigma_2^2}{E_2^0(1-d_2)}+\frac{\sigma_6^2}{G_{12}^0(1-d_6)}\right]=\frac{1}{2}\boldsymbol{\sigma}^{\mathrm{T}}\boldsymbol{C}_1\boldsymbol{\sigma} \tag{A1}$$

where $\boldsymbol{C}_1$ is the compliance matrix which takes the following form:

$$\boldsymbol{C}_1=\begin{bmatrix}\frac{1}{E_1^0(1-d_1)} & \frac{-v_{12}^0}{E_1^0} & 0\\ \frac{-v_{12}^0}{E_1^0} & \frac{1}{E_2^0(1-d_2)} & 0\\ 0 & 0 & \frac{1}{G_{12}^0(1-d_6)}\end{bmatrix} \tag{A2}$$

Three upper left determinants of the above symmetric matrix $\boldsymbol{C}_1$ are given by

$$\begin{cases}\varDelta_1=\frac{1}{E_1^0(1-d_1)}\\ \varDelta_2=\frac{1-v_{12}^{0,2}(1-d_1)(1-d_2)}{E_1^0E_2^0(1-d_1)(1-d_2)}\\ \varDelta_3=\frac{1-v_{12}^{0,2}(1-d_1)(1-d_2)}{E_1^0E_2^0G_{12}^0(1-d_1)(1-d_2)(1-d_6)}\end{cases} \tag{A3}$$

As $0\leqslant d_{1,2}<1$ and $v_{12}^0\approx 0.06<1\leqslant\frac{1}{(1-d_1)(1-d_2)}$, one can determine that $\Delta_1>0$, $\Delta_2>0$ and $\Delta_3>0$. The above determinant test proves that $\boldsymbol{C}_1$ is positive definite. This further verifies that the real quadratic form $\boldsymbol{\sigma}^{\mathrm{T}}\boldsymbol{C}_1\boldsymbol{\sigma}$ is positive, so is $\rho\psi_{\mathrm{e}}$ in Eq. (A1) in this stress state.

2. If $\sigma_1<0$, $\sigma_2<0$, the expression of $\rho\psi_{\mathrm{e}}$ in Eq. (2) reduces to

$$\rho\psi_{\mathrm{e}}=\frac{1}{2}\left[\frac{\varphi(\langle-\sigma_1\rangle,d_1)}{E_1^0}-\frac{2v_{12}^0\sigma_1\sigma_2}{E_1^0}+\frac{\varphi(\langle-\sigma_2\rangle,d_2)}{E_2^0}+\frac{\sigma_6^2}{G_{12}^0(1-d_6)}\right] \tag{A4}$$

where

$$\begin{aligned}\varphi(\langle-\sigma_1\rangle,d_1)&=\frac{1}{k_1^2}\left[\langle-k_1\sigma_1\rangle^2+\ln\left(1-d_1\mathrm{e}^{-\langle k_1\sigma_1\rangle^2}\right)-\ln(1-d_1)\right]\\ &\overset{\sigma_1\leq 0}{=}\frac{1}{k_1^2}\left[(k_1\sigma_1)^2+\ln\left(\frac{1-d_1\mathrm{e}^{-(k_1\sigma_1)^2}}{1-d_1}\right)\right]\end{aligned} \tag{A5}$$

in which

$$\ln\left(\frac{1-d_1\mathrm{e}^{-(k_1\sigma_1)^2}}{1-d_1}\right)>0 \tag{A6}$$

thus we can determine that

$$\varphi(\langle-\sigma_1\rangle,d_1)>\sigma_1^2\quad\text{if }\sigma_1<0 \tag{A7}$$

Similarly, the following inequation can be deduced:

$$\varphi(\langle-\sigma_2\rangle,d_2)>\sigma_2^2\quad\text{if }\sigma_2<0 \tag{A8}$$

So we can conclude that

$$\rho\psi_{\mathrm{e}}>\frac{1}{2}\left[\frac{\sigma_1^2}{E_1^0}-\frac{2v_{12}^0\sigma_1\sigma_2}{E_1^0}+\frac{\sigma_2^2}{E_2^0}+\frac{\sigma_6^2}{G_{12}^0(1-d_6)}\right]=\frac{1}{2}\boldsymbol{\sigma}^{\mathrm{T}}\boldsymbol{C}_2\boldsymbol{\sigma} \tag{A9}$$

where $\boldsymbol{C}_2$ is another compliance matrix that takes the form

$$\boldsymbol{C}_2=\begin{bmatrix}\frac{1}{E_1^0} & \frac{-v_{12}^0}{E_1^0} & 0\\ \frac{-v_{12}^0}{E_1^0} & \frac{1}{E_2^0} & 0\\ 0 & 0 & \frac{1}{G_{12}^0(1-d_6)}\end{bmatrix} \tag{A10}$$

It can be easily verified that all three upper left determinants of the matrix $\boldsymbol{C}_2$ are positive, therefore we can also determine from Eq. (A9) that $\rho\psi_{\mathrm{e}}>0$ under this loading condition.

3. If $\sigma_1>0$, $\sigma_2<0$, the corresponding expression of $\rho\psi_{\mathrm{e}}$ in Eq. (2) reduces to:

$$\rho\psi_{\mathrm{e}}=\frac{1}{2}\left[\frac{\sigma_1^2}{E_1^0(1-d_1)}+\frac{\varphi(\langle-\sigma_2\rangle,d_2)}{E_2^0}-\frac{2v_{12}^0\sigma_1\sigma_2}{E_1^0}+\frac{\sigma_6^2}{G_{12}^0(1-d_6)}\right] \tag{A11}$$

Substituting the inequality (A8) into the above equation, the following relation can be derived:

$$\rho\psi_{\mathrm{e}}>\frac{1}{2}\left[\frac{\sigma_1^2}{E_1^0(1-d_1)}+\frac{\sigma_2^2}{E_2^0}-\frac{2v_{12}^0\sigma_1\sigma_2}{E_1^0}+\frac{\sigma_6^2}{G_{12}^0(1-d_6)}\right]=\frac{1}{2}\boldsymbol{\sigma}^{\mathrm{T}}\boldsymbol{C}_3\boldsymbol{\sigma} \tag{A12}$$

where $\boldsymbol{C}_3$ is a compliance matrix:

$$\boldsymbol{C}_3=\begin{bmatrix}\frac{1}{E_1^0(1-d_1)} & \frac{-v_{12}^0}{E_1^0} & 0\\ \frac{-v_{12}^0}{E_1^0} & \frac{1}{E_2^0} & 0\\ 0 & 0 & \frac{1}{G_{12}^0(1-d_6)}\end{bmatrix} \tag{A13}$$

Again, one can determine that all the three upper left determinants of $\boldsymbol{C}_3$ are positive, and further deduce that $\rho\psi_{\mathrm{e}}$ in Eq. (A11) is positive.

4. If $\sigma_1<0$, $\sigma_2>0$, the corresponding expression of $\rho\psi_{\mathrm{e}}$ in Eq. (2) reduces to

$$\rho\psi_{\mathrm{e}}=\frac{1}{2}\left[\frac{\varphi(\langle-\sigma_1\rangle,d_1)}{E_1^0}+\frac{\sigma_2^2}{E_2^0(1-d_2)}-\frac{2v_{12}^0\sigma_1\sigma_2}{E_1^0}+\frac{\sigma_6^2}{G_{12}^0(1-d_6)}\right] \tag{A14}$$

The following relation is derived by substituting the inequality (A7) into the above equation:

$$\rho\psi_{\mathrm{e}}>\frac{1}{2}\left[\frac{\sigma_1^2}{E_1^0}+\frac{\sigma_2^2}{E_2^0(1-d_2)}-\frac{2v_{12}^0\sigma_1\sigma_2}{E_1^0}+\frac{\sigma_6^2}{G_{12}^0(1-d_6)}\right]=\frac{1}{2}\boldsymbol{\sigma}^{\mathrm{T}}\boldsymbol{C}_4\boldsymbol{\sigma} \tag{A15}$$

in which $\boldsymbol{C}_4$ takes the form

$$\boldsymbol{C}_4=\begin{bmatrix}\frac{1}{E_1^0} & \frac{-v_{12}^0}{E_1^0} & 0\\ \frac{-v_{12}^0}{E_1^0} & \frac{1}{E_2^0(1-d_2)} & 0\\ 0 & 0 & \frac{1}{G_{12}^0(1-d_6)}\end{bmatrix} \tag{A16}$$

Similarly, one can easily determine that $\rho\psi_{\mathrm{e}}$ in Eq. (A14) is positive under this loading condition by proving the positive definiteness of the symmetric matrix $\boldsymbol{C}_4$.

As a result, it can be concluded that the complementary elastic strain energy density $\rho\psi_{\mathrm{e}}$ proposed in Eq. (2) is positive definite for all nonzero in-plane stress states.

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